

## Problem sheet 5: Basic Quantum Information Protocols

**Problem 1 (HW). Entanglement swapping.** Suppose that Alice and Bob both share a maximally entangled state with Charlie. Describe the procedure by which Alice and Bob can generate a maximally entangled qubit pair by only acting on their local quantum system and exchanging classical bits.

**Problem 2. Error correction and Kraus operators.**

- (a) Suppose that an error correcting code  $V$  can correct errors from a noise channel with Kraus operators  $\{X_i\}$ . Show that it can also correct errors for any channel which has Kraus operators  $\{Y_j\}$ , where each  $Y_j$  is a linear combination of  $X_i$ 's.
- (b) Consider an error correcting code on a system  $B$  of  $n$  qubits, with  $P$  the projection operator on the code subspace. If  $M$  is a qubit operator, let  $M_i$  denote the operator which acts as  $M$  on the  $i$ -th qubit and as  $\mathbb{1}$  on all other. Suppose that  $PM_iN_jP$  is proportional to  $P$ , for all  $i, j = 1, \dots, n$  and  $M, N \in \{I, X, Y, Z\}$  are arbitrary Pauli operators. Show that we can correct an arbitrary error channel of single-qubit errors, i.e., any channel of the form

$$\Phi_B = \sum_{i=1}^n p_i \Phi_i,$$

where  $\{p_i\}$  is a probability distribution and  $\Phi_i$  is a channel which acts only on the  $i$ -th qubit.

**Problem 3. Remote state preparation.** This is about a protocol called *remote state preparation*, which is closely related to quantum teleportation, but here only one bit of classical communication is required to remotely prepare a given qubit state. Compared to the teleportation, the sender knows a classical description of the state to prepare, and has access to a larger number of entangled qubits.

- (a) Let  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \in \mathbb{C}^2$  be a pure qubit state. Show that

$$(|\psi\rangle\langle\psi|)^T = |\bar{\psi}\rangle\langle\bar{\psi}|, \quad \text{and} \quad \mathbb{1} - |\bar{\psi}\rangle\langle\bar{\psi}| = |\bar{\psi}^\perp\rangle\langle\bar{\psi}^\perp|,$$

where the transpose is with respect to the computational basis, and  $|\bar{\psi}\rangle = \bar{\alpha}|0\rangle + \bar{\beta}|1\rangle$ ,  $|\bar{\psi}^\perp\rangle = \bar{\beta}|0\rangle - \bar{\alpha}|1\rangle$ .

- (b) Suppose that Alice and Bob share a maximally entangled state  $|\Phi_{AB}^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . Alice would like to give Bob the state  $|\psi\rangle$ , but she does not want to give Bob any information about  $\alpha$  and  $\beta$ .

Alice performs a projective measurement on her part of the maximally entangled state, corresponding to the projectors  $\Pi_0 = |\bar{\psi}\rangle\langle\bar{\psi}|$  and  $\Pi_1 = \mathbb{1} - \Pi_0$ . Show that the outcome probabilities of this measurement are both  $1/2$ .

- (c) Alice then sends Bob the single-bit outcome of her measurement  $x$  in a classical system  $C$ . Show that Bob now holds the state

$$\rho_{BC} = \frac{1}{2}|\psi_B\rangle\langle\psi_B| \otimes |0_C\rangle\langle 0_C| + \frac{1}{2}|\psi_B^\perp\rangle\langle\psi_B^\perp| \otimes |1_C\rangle\langle 1_C|,$$

where  $|\psi^\perp\rangle = \beta|0\rangle - \alpha|1\rangle$ .

- (d) Assume, for the moment, that  $|\psi\rangle$  is of the form  $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$ . Show that

$$Z|\psi^\perp\rangle\langle\psi^\perp|Z = |\psi\rangle\langle\psi|,$$

and hence describe how Bob can recover the state  $|\psi\rangle$  from  $\rho_{BC}$ .

- (e) Suppose that Alice wants to send Bob many qubits  $|\psi_1\rangle, \dots, |\psi_n\rangle \in \mathbb{C}^2$ , which do not necessarily take the above form. Assume that Alice and Bob share  $n \cdot m$  maximally entangled states, where  $m = 2^{n+\log n}$ . They each arrange their qubits in a rectangle, so that the qubit from the  $(i, j)$ -th maximally entangled pair lies in the  $i$ -th row and  $j$ -th column. For each  $i$ ,  $i = 1, \dots, n$ , Alice measures the entire  $i$ -th row of qubits in the  $\{|\bar{\psi}_i\rangle, |\bar{\psi}_i^\perp\rangle\}$  basis. Show that, with high probability, there will be an entire column of qubits for which the measurements were successful, that is, the result corresponded to the projector  $\Pi_0 = |\bar{\psi}\rangle\langle\bar{\psi}|$ .
- (f) Alice sends Bob the index  $j = 1, \dots, m$  of such column classically. Show that in this way, Alice can remotely prepare  $n$  states in Bob's system with approximately 1 bit of classical communication per state, in the limit as  $n \rightarrow \infty$ .